

Photography Community Codesign (a.k.a. Codesign in a Global Pandemic)

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ABSTRACT

Codesign brings together users and creators of a product to collaboratively determine how that product will manifest. In this case study of codesign, the Adobe Lightroom team collaborated with 12 photographers in a 4-day workshop to envision an in-app community space for photographic expression and learning. We discuss findings from the codesign workshop, including core values for the envisioned photography community: authenticity, connection and growth, and how we translated these themes to the design of the product. This case study demonstrates a successful codesign effort within the context of user-centered design at a large company.

CCS CONCEPTS

• **Human-centered computing** → *Social networking sites; User studies; Participatory design.*

KEYWORDS

Community, Photography, Codesign, Design, Creativity, Representation, Safety, Learning, Inclusion

ACM Reference Format:

Jenna Frens, Lindsey Rasie, and Victoria Hollis. 2022. Photography Community Codesign (a.k.a. Codesign in a Global Pandemic). In *CHI '22: ACM Proceedings on Conference in Human-Computer Interaction*. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3491101.3503573>

1 INTRODUCTION

The photography community is a globe-spanning community of practice [8][15] that unfolds locally in person and online in network publics [4]. The term ‘online community’ colloquially means any digital space where people gather. Although this can be synonymous with social media platforms (e.g., Facebook, Instagram), within the photography community, this includes niche spaces within apps and websites (e.g., Flickr, VSCO, 500px). Not every digital image-sharing space is an online photography community. In the present study, we set out to clarify the concept of “community” with photographers, and how to embed positive values in the design of a digital photography community.

Lightroom is an industry-standard software that photographers use to edit and organize photos. The Lightroom team saw a unique

opportunity to build a photography community space within the tool, by facilitating peer-to-peer interactions and bringing people together to share information around photo-editing. We hope to bring a community experience to users, from new photographers to accomplished experts.

1.1 Motivations and Challenges

Other Adobe communities (e.g., Behance) are not embedded within image-editing tools. This new initiative raised questions for our team of researchers, designers, product managers, engineers and photography instructors.

Who do we want to be? Since the mid-1990s, Lightroom has been a photo-editing tool. In evolving this identity, our research aimed to uncover core values aligned with the photography community. It was important also to identify who we don’t want to be. Our value proposition should stand alone in an over-saturated world of digital spaces.

What mechanisms will exist to drive this community? Once we scope an identity, we need to align features, capabilities, and best practices to drive that vision to fruition. We worked to identify vital instruments and concepts for constructing community, while also infusing unique Lightroom technical capabilities (e.g., step-by-step edit videos).

How do we design a space for everyone? Photography technology has historically replicated discrimination against marginalized identities. For example, Kodak Shirley Cards served as a tool for color balancing and measuring skin tones starting in the 1950s, anchoring “whiteness” as the normative standard [11]. The Polaroid ID-2 camera, with its modified flash to photograph dark skin, was used during South African apartheid in the development of “Pass Books.” Pass books were identification booklets that recorded the movements of black South Africans in and out of “white zones,” and were critical to the continued maintenance of segregation [16].

A widespread, ongoing phenomenon in online communities is the amplification of hate and marginalization of minoritized identities. Online communities have helped racist individuals and groups spread hateful ideologies [3], and algorithms designed to curate online content have embedded discriminatory ideologies and learned from biased datasets, skewing web searches and tagging images in ways that further marginalize minoritized people [9]. Professor of African American studies Ruha Benjamin ties together the history of bias in photography technology with present-day digital discrimination in her discussion of *coded exposure*, where who is seen becomes coded directly into technology [2]. It is crucial to confront discriminatory bias in designing new technologies that risk replicating systemic hierarchies of oppression, by raising the

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ACM CHI '22, May 03–05, 2022, New Orleans, LA

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ACM ISBN 978-1-4503-9156-6/22/04.

<https://doi.org/10.1145/3491101.3503573>

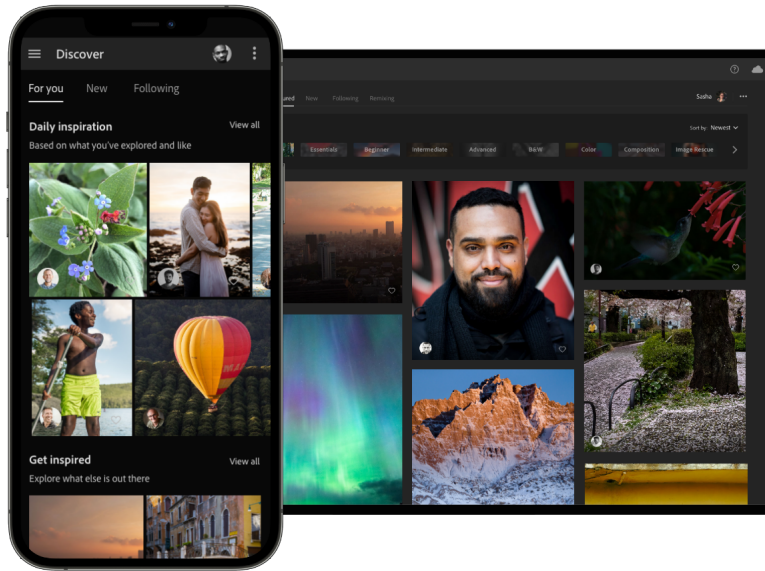


Figure 1: The Lightroom Discover community UI, reflecting changes from the codesign workshop.

stories and voices of marginalized people [10], especially for online communities.

We agreed from the start that we did not want to recreate patterns of marginalization towards non-male, non-cis, and non-white voices in our photography community space. We felt morally driven on this front, and furthermore hypothesized that an inclusive community space would benefit the business fiscally. We wished to build a space that would be beginner-friendly, multi-generational, and welcoming towards people of different lived experiences, photography subjects and styles. However, conflicting needs between different photographers may introduce design complexities. Our goal was to identify where those areas of conflict may arise and how they may be addressed.

How do we reconcile building a community within a corporation? We also recognized the inherent tension in building a free-to-use creative community within a publicly traded corporation. We had to grapple with how these could co-exist, navigating issues such as paywalls, moderation, and community ownership. User-empowered research would be critical to help us build a positive space that honors photographers and meet business needs.

1.2 Research Questions

- (1) How do photographers define "community"?
- (2) How do communities foster safety and inclusion?
- (3) What experience(s) do photographers want from the Lightroom community?
- (4) How do photographers envision the future design of Lightroom Learn and Discover?

1.3 Value of Codesign for Community Building

We define *codesign* as collaboration among the users and creators of a product, with the goal of empowering diverse stakeholders, especially those who have been underrepresented or excluded from

the design process, to influence product design. More simply, codesign means designing "with" instead of "for." Codesign is often synonymized with *participatory design*, a process of community-driven design rooted in Scandinavian political processes [12], that takes place within *user-centered design* [1]. Our role as researchers conducting codesign was as contributor-facilitators. Holding positions in a multi-billion dollar corporation, we are in highly privileged positions and cannot have the same experiences as all photographers. This difference of lived experience extends to intersections of ethnicity, race, culture, gender, ability and more. As such, our position in this workshop was primarily facilitation based, but our own experience was inserted at key inflection points through questions and moderation to generate empathy and understanding between participants. In this codesign process, responsibility falls on company stakeholders to decide whether and how to empower users by implementing their ideas. As researchers, we have a responsibility to continue the conversation and build accountability between the product team and community post-workshop.

Codesign generates design ideas, builds empathy, and engages communities, making the method particularly suitable to the development of Lightroom Learn and Discover, an in-app learning community. The codesign took place after the first public release of Lightroom Learn and Discover, with open opportunities to change existing designs, influence planned upcoming feature development, as well as co-build a future feature roadmap. We chose codesign because the best community spaces are those designed by the communities that they are for.

2 METHOD

2.1 Codesign Participants and Recruiting

2.1.1 The Photographers. We carefully recruited photographers to ensure representation of three community roles: leaders, mentors, and learners. Perspectives represented by participants in a codesign

workshop will directly influence the design outcome, so we selected for roughly equal representation among these roles.

- **Leaders (n=4)** were defined as photographers who had participated in leadership roles in photography communities, moderated online photography groups, or organized photography events.
- **Mentors (n=5)** were defined as experienced photographers with teaching or formal mentoring experience.
- **Learners (n=3)** were defined as photography students, aspiring professionals, or photography hobbyists.

Photographers were recruited through a snowball sampling method, starting with established Lightroom contacts. We asked contacts to refer photographers. We quickly found that this approach was skewing the participant group to be homogeneous (predominantly cisgender, male, and white). To recruit a more diverse group, we contacted three external organizations that elevate female, trans, nonbinary and POC photographers, and continued with snowball sampling.

Before extending an invitation, we interviewed potential participants about their interest in photography communities, availability to participate, and attitudes toward collaboration. This recruiting process unfolded throughout January and February, 2020. The codesign was to be an in-person event in New York City (NYC), so attending in-person was an original requirement. Photographers received one year of Adobe Creative Cloud and 700 USD for participating.

2.1.2 The Lightroom Team. The Lightroom team consisted of three researchers, three designers, two product managers and one engineer (n=9). All were full-time Adobe employees. In addition to being product domain experts, team members ranged from novice to expert in photography.

2.2 Workshop Procedure

2.2.1 Codesign and COVID-19. In late February and early March 2020, shortly before the workshop, internal communication at Adobe rocked our study, as well as the entire company. First corporate travel was restricted, and then, all US offices were closed. As a result of rapidly increasing COVID-19 cases in NYC, our study could not go on in-person as planned. We moved the study online, assessed any new participant needs, and offered options to participate partially or drop out from the study. All participants agreed to participate online.

2.2.2 Day 1: Storytelling and Interviews. Day 1 started with introductions. Each codesigner shared their name, pronouns, location, a little bit about themselves, and one value that they believed was important for a community. After introductions, we collaboratively set ground rules for speaking and listening during the workshop. This helped establish safety for open sharing. After that, we asked the photographers to share a photo that represented community to them. The Lightroom team equally participated. This session helped create a sense of connection and shared purpose.

After the introduction and storytelling sessions, we interviewed each photographer about their experiences with community for 60 minutes. The purpose of the interview was to understand what community means to photographers, and how to build a safe and

inclusive community. Photographers were more comfortable talking about emotionally-charged experiences in the 1:1 setting. This gave them practice before speaking in front of the rest of the group.

2.2.3 Day 2: Panels and Photo Challenge. We facilitated discussion panels where photographers took turns being panelists and audience. The three 75-minute (with breaks) panels were segmented by role: learners, mentors and leaders, with different questions for each role. The audience, consisting of the Lightroom team and photographers, watched and asked additional questions. The purpose of each panel was to continue to address research questions 1 and 2, ensuring that each perspective was shared with the rest of the group ahead of codesign breakouts. After the panels, we hosted an optional activity: the mobile photography challenge. The purpose of this was to build empathy for mobile-only users, do an online photographic critique session, inform design, and to have fun.

2.2.4 Day 3: Codesign Breakouts and Concept Feedback. We facilitated codesign breakout groups. In each group, there was one learner, one mentor, and one leader, matched with a designer, project manager or engineer, and researcher from the Lightroom team. The breakout groups each focused on corresponding design questions: How do we envision learning and tutorials? How do we envision sharing and inspiration? How do we envision interaction and community? In the 120-minute (with breaks) session, we jumped right into codesign in the literal sense of collaboratively designing together. We posed a few questions to get people started, and let the conversation evolve from there. The designer in each group sketched out ideas in collaboration with the other codesigners using a shared digital Miro board, and the researcher moderated the discussion to ensure everyone had opportunities to speak.

After the codesign breakout session, there was a feedback panel for an upcoming June product release. The Lightroom team shared a presentation and demo, answered questions, and solicited feedback. This gave photographers a chance to see what we were already working on prior to the codesign, give feedback directly to the design and product team, and have near-term influence on ongoing product development in addition to helping set the overall vision.

2.2.5 Day 4: Community Guidelines and More Codesign Breakouts. We conducted a 75-minute (with a break) community guidelines session and continued the codesign breakouts. The central purpose of the community guidelines session was to center on community values, and to co-write community guidelines. Each person had an opportunity to write a community mission statement. We also discussed how the Lightroom team should respond if someone breaks the rules. After the session ended and another break, codesigners returned to their breakout groups from the previous day to continue collaborating on design sketches for another 60 minutes. The day ended with a closing session where each participant shared their final, most important piece of advice to Lightroom. We dismissed with promises to share updates at later dates.

2.3 Codesign Principles

- **Put empathy first.** Recognizing that codesigners were sheltering in place during a crisis, we ensured everyone felt like they were being treated with humanity. For instance, letting people know that they could go off-camera, miss sessions, or

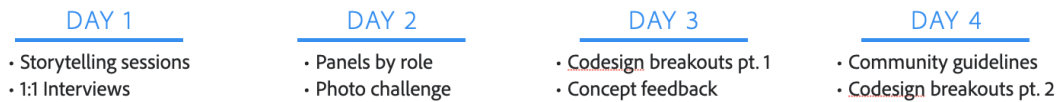


Figure 2: Codesign workshop schedule.

withdraw from the study without compensation penalization. We also kicked off the workshop by collectively establishing ground rules so everyone would feel comfortable participating.

- **Encourage diverse perspectives.** Beyond recruiting diverse perspectives, we encouraged reflection before public sharing. To ground each photographer, we interviewed them 1:1 about their experiences of community and ideas for ours. Next, we grouped leaders, mentors, and learners together to uncover commonalities and shared ideas among those with similar roles. Finally, design breakout groups would each have one leader, one mentor and one learner, so diverse perspectives would influence each design artifact produced in the codesign. We started with individual stories, moved to similar role sharing, and then cross-functional codesign groups.
- **Build connection throughout.** Codesigning a community should feel like being part of a community. To foster a sense of connection, we began the workshop with a relationship building exercise. Over the week, we ensured that there was time built in for informal chatting, especially in smaller breakout groups. At the end of the workshop, we created space and time for people to share contact information if desired, and to say goodbye.

3 ANALYSIS AND FINDINGS

Qualitative analyses were conducted bottom-up across individual participant interviews, panels, and co-design activities. Each activity was first analyzed by the respective researcher who moderated the session. Through group review, smaller themes were organized into broader themes resulting in a framework of high-level desired characteristics of an online photography community and specific ways to support those characteristics. Quotes in the below findings are from the photographer participants, who each gave explicit permission for their statements to be published.

3.1 ‘Community’ Means a Shared Purpose or Identity. It’s Not Just Content or a Platform

‘Communities’ were differentiated from other social groups based on a central ‘why’ for membership: a shared purpose or identity that tied members together. For example, meetups to critique street photography (shared purpose) or women in photography (shared identity).

“The difference between a group and community is the difference between a house and a home... When you ride the train to work with 100 people, that is a group. But when you arrive to work with 100 people that have

the same purpose, and a culture emerges, that becomes a community.” P3, Leader

Referring to prior experiences, participants desired communities that allowed them to feel **authentic**, supported **personal growth** and enabled **human connection**. These characteristics often intersected with and benefited one another. For example, being authentic allows for greater experimentation which fosters growth in a creative skill. We saw converging evidence on the importance of these desired characteristics, which we describe further below.

“You feel like you can express yourself and do things you normally wouldn’t do...If people aren’t going to judge you, then you tend to experiment more and find yourself more... The freedom to create.” - P4, Leader

3.2 Authenticity is Fostered by Psychological Safety and Avenues for Self-Representation

Being authentic and vulnerable with community members was considered a critical community attribute. This allows photographers to share their ups and downs without as much emphasis on careful self-presentation [14][5][13]. Photographers contrasted this against online spaces where users must carefully tailor self-presentation due to concerns about being evaluated by peers and potential employers (e.g., public portfolios or social media).

Enabling authenticity is complex. Foremost, participants must be able to see themselves as part of a pre-existing community through **representation** and **inclusion** of others like them. For a creative online community centered on photography, that meant being able to see **diversity** in skill sets, art styles, and identities. A first impression of lacking representation can cause potential members to self-select out of the community.

“I never saw anyone like me as a photographer [growing up]... It’s about representation. If you’re not able to see it, you can’t imagine it.” - P6, Mentor

Participants must feel **safe** that no harm would befall them for expressing their identities or differences in skill, so long as that expression of identity does not conflict with community guidelines.

“Sense of belonging, sense of feeling embraced and it’s a safe place for you to be you” - P1, Leader

Ideas arose on how to build a community with positive first impressions of safety and authenticity.

3.2.1 Build Safety: Create a Clear Mission Statement and Community Rules. Safety starts with setting guidelines and tone when first entering the community. This includes a mission statement of what the community is for, explicit rules on what is acceptable, and how rules are enforced. Participants wanted to feel a sense

of co-ownership over the community. This meant they could self-enforce community guidelines and provide feedback to Adobe about concerns.

Workshop participants co-created a mission statement and community guidelines that emphasized being constructive and respectful with critiques, and empowered users to tell their stories: “A safe, inclusive and supportive community with a diverse set of skills, interests, and identities that is dedicated to learning and sharing photographic expression”. The guidelines also outlined clear consequences for unacceptable behaviors (hate speech, copyright violation, doxxing, etc.).

3.2.2 Foster Inclusivity: Create a Welcoming First Experience. Codesigners stressed the importance of inclusion of different skill levels and photography backgrounds, as well as the encouragement of newcomer participation.

- Onboard new users with an activity they can immediately contribute towards regardless of their skill level
- Teach the language of photography which can otherwise be isolating for people who are new to the field
- Encourage responses to first posts so contributions feel valued rather than like speaking into a void

“It would be really sad...[if] someone posts a photo and doesn't get any feedback. To not get any movement on a photograph is really hard. Otherwise why else would you be doing it?” - P6, Mentor

3.2.3 Showcase Diversity: Spotlight New Contributors, Not Just Professionals. Throughout the workshop, participants highlighted not wanting a homogeneous community. Photographers are not just diverse in skill levels, but also in their art styles, the subject matter of their photography, their personal identities, processes and equipment, and geography.

“Entering a space and being like ‘No one looks like me, huh.’ How can you not take that personally or how can you not feel like, ‘Oh, so this place wasn't made for me.’ That's what you think.” - P2, Leader

In codesign exercises, the design concepts for photo feeds highlighted contributions from around the world, different aspects of the photography process (not just end results), photos from different skill levels, as well as hashtags and filtering that would allow users to narrow on particular interests (e.g., astrophotography time-lapses). Participants specifically suggested having a separate space for new photographers to be featured, rather than relying on feeds where only trending photographers are showcased.

3.2.4 Enable Self-Representation: Create Space Where the Community has Control. Participants described the importance of being able to represent themselves rather than a corporation representing them. For codesigners this meant community profile control, affordances for letting community members build their own niche groups for shared interests, and letting community members have leadership opportunities to create or moderate groups.

“I have seen so many places where I feel like I was represented by someone who I don't agree with, but then people just lump me to that community. I just don't want to be represented anymore. I think that's

a shared sentiment by a lot of people who have identities on the margins.” - P1, Leader

They clarified: *“I don't want to be represented in a hasty way. I still want to see authentic and nuanced representations.”*

3.3 Growth Within a Community Requires Supporting Functional and Emotional Needs for Learning

Communities centering on personal growth was important for participants. Codesigners highlighted community characteristics that support both functional and emotional needs for learning.

“If you give up on learning, you've given up on growing as a creative. Learning is investing in yourself and is important at any level.” - P4, Leader

3.3.1 Functional Needs: Enable Sharing of Credible and Appropriate Resources.

- **Credible information.** Participants wanted assurances that the information they're acting on was truthful and current. This can be a benefit of formal education and a downside of online spaces. Participants reported finding outdated or purposefully false information on the internet, hindering their ability to grow. Centralized resources provided by accredited industry experts would be a large community benefit to participants.
- **Connection of their goals to appropriate learning resources.** The photography space has decades of professional terminology built into it. Participants stressed difficulty for newer photographers when translating creative goals into industry language, leading to an inability to locate needed resources. Participants felt that learning resources (and navigation of those resources) should speak to users at all levels.

“Why are they putting fake information out there? It's hard to trust what they're saying. I have to put time and effort on what they are talking about.” - P11, Learner

“We all need different information, so we need to have a structure to help people on a different level...” - P4, Leader

3.3.2 Emotional Needs: Build Confidence, Not Competition. Negative community interactions could exacerbate an already difficult and intimidating experience for those new to a field, while positive interactions could encourage photographers and help prevent disengagement due to low confidence [6]. While competitions were common in photography communities, participants warned that they can pit members against one another demotivating participation and hurting relationship building.

“Having other people that say you gotta try, you gotta push yourself, you have to do that thing that scares you.” - P12, Learner

Even if it's a small thing, I always have this pressure...When we think of 1st, 2nd, or 3rd, we think this is the best photo...this isn't encouraging people to come up with ideas. They are trying to fight...” - P11, Learner

3.4 Meaningful Connection is Achieved by Enabling Communication and Identity Signaling

Co-designers described how they wanted to have communication in a photography community, including a space to gather critiques and impressions of their creative processes. Dialogue was necessary for the space to feel like community.

“...if you want to build community, it’s kind of fundamental to have discussion threads.” - P8, Mentor

Codesigners desired a steady stream of user-generated activity where members are motivated to give back rather than a 1-sided conversation where only Adobe generates content. Members needed to showcase their voices.

3.4.1 Enable Rich Communication: Free-Form Conversation is Key for Critique. Critique spaces were desirable, but participants warned that how feedback is delivered in those spaces must balance functional and emotional learning needs (as described above). The feedback must be actionable, yet not overwhelming to the extent that it discourages goal striving. This is reflective of findings from psychological theories of goal-striving [6].

- **Requesting critique:** When requesting critique, participants felt it was important to provide additional context for the reviewer. The poster should be explicit in the type of feedback desired and signal their background and skill level, so the feedback could be tailored specifically to the poster’s relative experience, for example, ensuring a beginner gets suitable feedback for what they can accomplish compared to a professional photographer. We saw examples of such behavior in the critique activity of the workshop, and these preferences actualized in the community co-design activities.

“...they understood and appreciated the goal of the image as well as suggest what might improve it.” - P5, Mentor, on describing ideal critique feedback]

- **Giving critique:** Appropriately, more advanced photographers in the codesign echoed the importance of speaking to the posters’ goals, as well as tailoring feedback so it’s actionable, but achievable. These findings refuted the assumption that simply posting images to the community would be suitable for feedback and showed how open-text communication would be necessary to serve this use case appropriately.

“Rarely do I say ‘I do not like.’ I talk about what’s effective and ineffective and check what they are trying to do.” - P9, Mentor

3.4.2 Build Identities: Let Users Signal Expertise, Interests and Desired Interactions. Stated previously, participants said self-representation is key to foster a safe, learning environment. In extension, users should have control over their public-facing profile. Codesigners wanted the capability to create a unique community identity, separate from their real-world identity, which could be uncomfortable to share online when being vulnerable (such as when learning).

“...It’s a little bit scary to show the very vulnerable...when I know it’s open to the public...Editors might look at it

*and think oh s*** they don’t know what they are doing...having very clear privacy settings I can play with.” - P1, Leader*

However, photographers described in codesign sessions how they would be interested in seeing information about others’ expertise, credibility, interests and other affinities [7], and desired interactions, even without a full name exposed. Community members would choose what and how much to share about themselves publicly. Participants prioritized certain profile elements: (1) Details for self-expression in terms of a bio, interest tags, and photographic styles. (2) Sample work which can signal their credibility to other members. (3) Work-in-progress pieces where they can optionally request feedback. They highlighted the importance of this space being distinct from completed work. (4) Opt-in ability indicating what types of relationships they are interested in having with other community members (e.g, badges to indicate they are a ‘mentor’ or ‘open to feedback’).

“Let’s say I’m following a photographer I really love and they are open to being a reviewer, I want to see if they are open to review my work...Having in their profile if they are willing to do things like that, that would be really cool.” - P7, Mentor

3.5 Outcomes: Immediate Changes, Further Guidance, and Process Advocacy

The codesign reinforced the idea that Lightroom should not emulate existing social media platforms. Instead, the team should lean into Lightroom’s unique capabilities as an editing suite that is part of a creative process. While other photography platforms are output-focused, Lightroom has an opportunity to be process-focused. There is also a unique opportunity to connect people together meaningfully over the common interests that drove users to the product initially. To build outwards and support these larger goals, the community space will need to specifically focus on fostering authenticity, growth, and connection amongst members.

Based on this research, the Lightroom team made immediate changes to community design concepts and its feature roadmap. The team reverted plans for incorporating social currency (e.g., like counts and follow counts in profiles), built space for new users to showcase their work (the ‘New’ feed), placed co-created community guidelines at the entry point of the community, wrote transparent messaging on where/how posted work will be showcased and used, and developed a system to showcase a more diverse set of featured subjects and photographers.

We have also reconvened codesign members at multiple junctions to provide further guidance, as well as ensure features and designs align to community pillars. In an additional effort to ensure Lightroom continues to deliver on its strategic priorities, Lightroom will launch a quantitative community health assessment to track how well the community supports authenticity, growth, and connection. This health assessment aims to keep Adobe and Lightroom accountable to their user base on the premise of equity, diversity and inclusion, the decided ethos of the community space.

This research also advocated for the codesign method throughout Adobe. Being able to leverage this method both live and remote

allows for broader employee and user participation and representation, which has made codesign a far more accessible and applicable method of research.

While the codesign process continues to evolve and improve as we encounter new user and product development challenges, at its core it remains a valuable tool to raise the voices of those from minoritized and marginalized groups. There is still much work to do to ensure the Lightroom community meets its intended vision, but without those user voices in the room to challenge team assumptions at the start, the Lightroom community would have potentially been on a different development path, been built on the wrong core values and unintentionally alienated the many types of photography enthusiasts - and people - that exist in the world today.

4 CONCLUSION

Codesign created a powerful moment of connection between photographers and the Lightroom team. It started a conversation that will be the throughline of future feature development and community evolution. It made the team think differently about the fundamental value a photography community should provide (growth rather than competition). It made the team think differently about diversity and inclusion. And it made the team think differently about what it means to build a meaningful community within a multi-billion dollar corporation. The road forward, while clearer, has not yet been paved in stone. Ensuring Lightroom continues to successfully navigate the tensions that exist between community and capitalism will require team alignment, methods of accountability, and continued touch points with the ever-changing user base. We must evolve as our users evolve; we must evolve together because the only way for Lightroom to build a successful community is as a community.

ACKNOWLEDGMENTS

Codesign only works when you have the belief and support of every single participant, Adobe and non-Adobe. We would like to thank the codesigners for sharing their time and expertise. We would also like to thank the Lightroom team for their enthusiasm to participate in the workshop and, ultimately, their willingness to translate insight into action. Special thank you to Natalie Be'er and Michael Cragg for lending their design expertise for visuals in this paper.

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